Samuel Oluyemi

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Software Developer/ Engineer

Software Engineer with strong foundation in OOP, data structures and algorithms, and operating systems (including embedded systems), and 3+ years of hands-on experience building web and mobile applications using JavaScript, React, Node.js, and Python. Strong background in developing scalable full-stack applications, REST APIs, database-driven systems, authentication, access control, cloud deployment, responsive interactive application features, as well as additional experience building Linux-based virtual machine infrastructure, AI-powered applications, and working with embedded software. Proven ability to collaborate in agile development teams, debug and troubleshoot software issues, and deliver user-focused solutions. Passionate about continuous learning, building reliable, scalable software, applying modern technologies to solve real-world problems, and growing focus on secure App development, systems and cloud engineering, and information security.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, SQL, TypeScript, C, C++
Frontend & Backend: React, React Native, HTML5, CSS3 & Node.js, Express.js, FastAPI, Flask, REST APIs
Databases: PostgreSQL, MySQL, SQLite, Flask-SQLAlchemy and adaptable to any other relational database
Web Technologies and Security: Authentication, Authorization, Access Control, API Rate Limiting, CRUD operations, Database integration
Systems: Linux, Ubuntu, virtual machines, networking fundamentals, embedded systems,
Dev Tools, DevOps, & Platforms: Git, GitHub, Figma, Adobe XD, Docker, Render, Vercel, Neon
Core Concepts: OOP, Data structures and Algorithms, Debugging, API integration, Concurrency fundamentals, Agile development, Multithreading

PROJECTS

Marketing & Advertising Platform – Work experience with Pill4rs Digital Company
Tech Blog World — Full-Stack Web Application (*HarvardX CS50 Final Project)*

- Developed a full-stack technology blog application that allows users to register, log in, read articles, comment on posts, and manage their accounts.
- Implemented authentication and session management using Flask-Login and Flask-WTF.
- Developed administrator functionality for creating, publishing, editing, and managing articles.
- Implemented role-based authorization to distinguish administrative privileges from regular user permissions.
- Designed database models for users, articles, comments, and other application data using Flask-SQLAlchemy and SQLite.
- Configured Docker support and deployed the application through Render.

Full-Stack Developer Portfolio Platform

- Designed and developed a responsive full-stack portfolio application using React and TypeScript.
- Developed backend REST API endpoints using Node.js and Express.js to support application functionality.
- Integrated PostgreSQL for persistent storage of portfolio content and application data.
- Implemented administrator authentication and protected API routes to restrict unauthorized access.
- Configured API rate limiting and server-side security middleware to strengthen application protection.
- Managed environment variables and production database configuration.
- Deployed the frontend and backend to separate cloud hosting platforms and integrated a cloud-hosted PostgreSQL database.
- Used Git and GitHub for source control and project version management.

Mobile App (React Native) – Zero to Mastery Course Project:

- Built a cross-platform mobile app using React Native
- Implemented navigation, reusable components, and API integration
- Focused on performance optimization and responsive UI

Retrieval-Augmented Generation (RAG) System

- Built a document question-answering system using FastAPI, FAISS, and OpenAI APIs.
- Implemented document ingestion, vector embeddings, semantic search, and contextual question answering.
- Designed the workflow to retrieve relevant document content before generating responses.

Wonderland Adventure Game | C++ (Athabasca University Course Final Project)

- Designed and developed a modular, text-based adventure game using C++17 and object-oriented programming, organizing game functionality across classes and header files.
- Applied core programming concepts including STL vectors, structures, enumerations, templates, pointers, and references to manage game entities, inventory, navigation, and state.
- Implemented file-based content loading and parsing, interactive command processing, input validation, randomized riddles and jokes, and player progression with lives and scoring.
- Implemented a timed pause/resume mechanism using C++ concurrency features, including threads, atomic variables, and chrono utilities.

C++ Data Structures and Algorithms

- Implemented and analyzed fundamental data structures and algorithms through university coursework.
- Worked with linked lists, binary search trees, balanced trees, hash tables, heaps, sorting algorithms, and graph traversal.

PROFESSIONAL EXPERIENCE

Technical Support Advisor (IT)
Concentrix Canada | Sep 2024 – Present

- Troubleshoot software and system issues across iOS, macOS, and Windows
- Support **25–35 users daily**, improving issue resolution efficiency
- Strengthened debugging, problem-solving, and communication skills

Software developer
Pill4rs Digital Company Ltd | Jan 2024 – Dec 2024

- Developed and maintained full-stack web applications (with 0ver 5000 users) using React, Node.js, and SQL
- Built scalable APIs and optimized database performance
- Translated UI/UX designs (Figma/Adobe XD) into responsive components
- Increased platform usage by **20%**, contributing to revenue growth

Web & Mobile Developer Intern
SolarSoft | Jul 2021 – Jul 2022

- Built and maintained applications using React, Node.js, and Python
- Assisted in debugging and improving application performance

IT Aide & Team Lead
Paris 2024 Olympics (IOC) | Jul 2024 – Sep 2024

- Provided technical support for **10,000+ athletes**
- Coordinated with cross-functional teams to ensure smooth operations

EDUCATION

Master's Degree – University of Alabama | 2024.
Bachelor's degree – University of Benin | 2018
Relevant Computer Science Coursework — Athabasca University: Introduction to Programming in C++; Data Structures and Algorithms; Operating Systems
Certifications: HarvardX – Introduction to Computer Science**;** FreeCodeCamp – JavaScript Algorithms & Data Structures; Responsive Web Design
Additional
Languages: English, French, Yoruba (Fluent) | German (Intermediate).
Interests: Software Engineering, Cloud Computing, Information Security, Data Privacy, Systems Infrastructure, and Technology Innovation.